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Viscoelasticity of non-colloidal hydrogel particle suspensions at the liquid–solid transition†

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Suspensions of soft particles transition from a viscous fluid to a soft material upon increases in phase volume. The criteria defining the transition to this jammed state are difficult to define due to the porous and deformable nature of soft particles. Here, we characterise the rheology of aqueous suspensions of industrially relevant non-colloidal, polydisperse, frictional agarose microgels and evaluate shear and viscoelastic behaviour across a range of phase volumes from the dilute regime to the highly concentrated regime. In order to model the viscoelastic response of suspensions without free fitting parameters, the random close packing volume fraction (ϕ_{rcp}) and the particle modulus are determined, respectively, from particle size distribution measurements and direct measurements of reduced elastic modulus of individual particles (E_{rp}) using Atomic Force Microscopy. It is found that at ϕ_{rcp} , previously shown to correspond to divergence of the viscosity, also corresponds to the suspension transition from a viscous to viscoelastic fluid. However, the transition to a jammed solid-like state (ϕ_j) occurs at phase volumes exceeding this value (*i.e.* $\phi_j > \phi_{\text{rcp}}$). The suspension modulus and its sudden growth at ϕ_j are well-predicted by the Evans and Lips model that incorporates the E_{rp} of the hydrogel particles. This rheological behaviour showing a dual transition is reminiscent of two families of systems: (i) colloidal suspensions and (ii) frictional-adhesive non-colloidal suspensions. However, it does not strictly follow either case. We propose that the width of the transition region is dictated by frictional contact, particle size distribution and particle modulus, and plan to further probe this in future work.

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1. Introduction

The liquid–solid transition for a disordered dispersion of repulsive particles is defined as the point at which it “loses the ability to flow whilst remaining amorphous”.¹ At the limit of zero-shear, theoretical hard sphere models predict jamming to occur at a discrete phase volume (ϕ_j), which should coincide with asymptotes in viscosity and shear elastic modulus.^{2,3} This transition is also suggested to occur at the random close packing fraction (ϕ_{rcp}) for purely repulsive (hard) monodisperse spheres,^{2,4} whilst repulsive, colloidal soft particles are shown to undergo dual transition at the glass and jamming fraction respectively.^{5,6} There is still little experimental data probing the nature of the microstructural state required for the liquid–solid transition (LST) for densely packed polydisperse,

frictional, granular soft particles. For soft interaction potentials and deformable spheres, the critical jamming transition is not necessarily so well-defined structurally because particles may deform in close proximity⁷ that allows a fluid-like response to be observed beyond ϕ_{rcp} , and thus the expected asymptotes in viscosity and modulus may not coincide with each other. This makes development of structure–rheology relationships challenging for suspensions of soft and deformable particles, which are ubiquitous throughout industry that includes coatings, foods, personal care and pharmaceutical products. Models have recently been proposed suggesting that, at and above jamming, elastic forces between particles and contact area increases with decreasing particle modulus.^{5,8–10}

The maximum packing fraction (ϕ_m) is defined in rheology as the critical phase volume at which the viscosity of suspensions asymptotes towards infinity. We recently showed that the value of ϕ_{rcp} , when obtained using the one-dimensional mapping model of Farr and Groot¹¹ and the measured particle size distribution, corroborates with ϕ_m for many colloidal and non-colloidal hard spheres.^{12,13} For suspensions without significant attractive forces between particles, the viscosity (η) is accurately described by the modified Maron–Pierce–Quemada

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(MPQ) model up to ϕ_{rcp} :

$$\eta = \eta_s \left(1 - \frac{\phi}{\phi_{\text{rcp}}} \right)^{-2} \quad (1)$$

where η_s is solvent viscosity and ϕ is disperse phase volume. The importance of the modified-MPQ model is that it provides a prediction of the viscosity based only on the independently measured particle size distribution and requires no arbitrary fitting to an empirical equation. Accurate prediction of experimental data using the model is an indicator for the absence of attractive interparticle forces and/or measurement artefacts, and we suggest it also provides greater certainty over the phase volume that is ill-defined for soft spherical particles that includes those that are swellable and/or porous.

For hydrogels, which are soft spheres that contain solvent within their structure, the phase volume is not easily determined and estimating phase volume using density, freezing transition¹⁴ or maximum packing fraction of 0.64 for a centrifuged suspension¹⁵ are not directly applicable. Hydrogel particles are swollen cross-linked polymeric particles and their specific volume can alter in response to changes in solution environment (*e.g.* osmotic and temperature induced swelling^{16,17}), compressive forces, and limited swelling due to confinement or a lack of solvent.¹⁸ These effects make structure-rheology relationships and the origin for specific transitions in response to variations in system parameters challenging to interpret for soft particle suspensions. For example, experimental results of Adams *et al.*¹⁹ and Di Lorenzo and Seiffert²⁰ suggest that the phase volume for a liquid–solid transition (LST) to occur is dependent on the particle modulus (G_p). We define this ‘jamming’ transition (ϕ_j) to be the phase volume corresponding to when G' increases above G'' . Adams *et al.*¹⁹ reports experiments showing that ϕ_j decreases with increasing G_p for suspensions of hydrogel microspheres consisting of agar. However, they highlight their uncertainty in the observation because of differences in particle size distribution and possible over-estimation of phase volume that was obtained using dry weight calculations. In the work of Di Lorenzo and Seiffert,²⁰ the particle phase volume was altered by osmotic deswelling of the particles; however, controlling phase volume in this manner also alters particle modulus, and thus the two parameters are convoluted. A thorough examination of LST for monodisperse soft colloidal microgels is described by Pellet and Cloitre.⁶ They show a transition region between the glass (Φ_g) and jamming transitions in which the suspension is a viscoelastic solid. While our suspension appears similar, the key differences in particle size, frictional contact and polydispersity suggest that the glass transition, and thus a viscoelastic LST, is not applicable here. Consequently, uncertainty persists in the literature concerning the relationship between the values of ϕ_m , ϕ_j and ϕ_{rcp} for soft particle suspensions, and the role of micromechanical properties of the suspension, such as G_p , in defining them.

Our recent study¹³ defined specific volume of microgels (in the form of hydrogel microspheres) based on dilute solution viscosity measurements utilising the Einstein and Batchelor equations; care was needed to ensure specific volume of the

microgels did not alter under compressive loads experienced during centrifugation that was used in their preparation. When defined in such a manner, we found that the predicted ϕ_{rcp} using the model of Farr and Groot¹¹ corroborated with the value of ϕ_m , which is the phase volume corresponding to asymptotic increase in measured viscosity. As a result of this approach, we discovered that the observed (apparent) dependencies of ϕ_m on G_p was in fact due to differences in ϕ_{rcp} that arose due to variations in the particle size distribution. This indicated that the modified-MPQ model accurately predicts the viscosity of deformable spheres up to ϕ_{rcp} when there are no significant attractive interactions (*i.e.* $\phi_m = \phi_{\text{rcp}}$) and demonstrated its important use in the analysis of such systems.

Here, we extend this previous work to volume fractions above ϕ_{rcp} by characterising the evolution of the suspension modulus with increasing phase volume to probe the liquid-to-solid transition (LST) and viscoelastic solid-like behaviour of these polydisperse, frictional hydrogel particles. The mechanical properties of the microgels are varied using polymer concentration. In contrast to previous works that defined the G_p based on the rheology of bulk gels at equivalent polymer concentration or based on the rheology of bulk suspension, we directly characterise the mechanical properties of individual hydrogel microspheres by performing nanoindentation using atomic force microscopy (AFM). The goal is to quantify the rheology during the LST and gain insight into the values of ϕ_m , ϕ_{rcp} and ϕ_j for suspensions of deformable spheres with consideration of their measured micromechanical properties. This is required for establishing a fundamental basis for developing understanding on the structure and rheology of colloidal microgel suspensions and more complex soft particle systems of industrial relevance. Clear definition of the nature of LST is relevant in fields such as: bioscience, for example cell colonies and tissues where solid-like behaviour can promote colony formation or modify tissue micromechanics;^{21,22} food technology where shelf stability and mouth-feel are influenced by the concentration-driven solid–liquid transition;²³ and pharmaceutical gels and creams where this transition can influence drug release.^{24–26} Knowledge of the drivers for a single discrete LST at random close packing ($f(\phi_0$ and PSD)), or two distinct transitions at random close packing and jamming ($f(G_p)$) will enable careful system design to prevent unstable and undesirable behaviour in these complex systems.

2. Experimental

2.1 Microgel manufacture

Agarose microgels were manufactured by forming water-in-oil emulsions and gelling the water droplets to form spherical microgel particles, as previously described.¹³ Microgels were washed with ethanol and centrifuged prior to dispersing in ultrapure water with resistivity of 18.2 M Ω cm (Sartorius Stedium). Microgels with concentrations of 1.5 wt% and 5 wt% agarose were made and dispersed in water at a range

of phase volumes. Dilution to the required phase volume was carried out immediately prior to measurement.

2.2 Rheology

Rheological measurements were carried out on the Haake Mars III rheometer, as described previously¹³ using 35 mm diameter parallel plates with smooth surfaces for $\phi < 0.2$; and roughened using sandpaper for suspensions with $\phi > 0.2$. A pre-shear was performed at 10 s^{-1} for 60 s to normalise shear history. Shear rheology measurements were performed with 30 s per point across a shear stress range from 0.1 Pa up to the measurable maximum stress, void of secondary flow effects. To determine the storage modulus (G') and loss modulus (G''), an oscillatory frequency sweep was measured within the linear viscoelastic region (LVR) at a stress of 10 Pa.

The modulus of the gel disks (at equivalent agarose concentrations as that in the microgels) was determined using parallel plate geometry as an approximation for particle modulus, G_p . Agarose solutions were prepared at 80°C and poured onto the non-slip (sandpaper) plate surface pre-heated to 80°C . The plates were brought together to the required gap, the sample carefully trimmed, and the temperature set point reduced to 25°C to form a gel. The gels were allowed 5 minutes standing time after the temperature set point was reached. An amplitude sweep was run to determine the LVR at high and low frequency followed by a frequency sweep across this range. G_p was estimated as the average value for G' in the LVR.

2.3 Particle size distribution and RCP

Particle size measurements were performed on the microgels using low angle laser light scattering with a Malvern Mastersizer 2000. This technique passes a laser light at a wavelength of 600 nm through the sample and measures the angle of the scattered light on a photosensitive detector. The particle volume is then calculated using Mie theory, which requires the refractive index of both the solvent and particle to be known.^{27,28} Particles were modelled as spheres and a variable refractive index was used dependent upon agarose concentration, according to Jain *et al.*,²⁹ who found a refractive index of 1.334 for 1.5 wt% agarose gel. Refractive indices for 5 wt% agarose microgels was calculated to be 1.342. The Mastersizer measurements were used to visually compare particle size distributions as well as comparing numerical values, with results showing a broad size distribution (σ) with average ($d_{0.5}$), surface weighted ($d_{4.3}$) and volume weighted ($d_{3.2}$) diameter (μm) shown in Table 1. Microgel effective phase volume (ϕ_0) was determined from dilute solution viscosity measurements, as described in detail in Shewan and Stokes,¹³

by adjusting specific volume (k) to fit Batchelor's model with the results shown in Table 1.

Random close packing fraction (ϕ_{rcp}) is determined using the analytical model of Farr and Groot.¹¹ Farr and Groot¹¹ developed a numerical model for predicting ϕ_{rcp} , and followed this with an analytical model comparable to their full numerical simulations with results output in a shorter time with little computing power. The analytical model works by mapping the 3D particle size distribution onto a set of rods in 1D, and allowing the rods of length L_i and L_j to interact through the potential function described by eqn (2). Here, h represent the gap between the rod ends and $f > 0$ is an adjustable free volume parameter. Thus, the 3D simulation of a polydisperse system of spheres becomes a much simpler 1D simulation of a polydisperse system of rods. To predict ϕ_{rcp} , the Farr and Groot analytical model utilises the log normal particle size distribution from the volume weighted ($d_{4.3}$) and surface weighted ($d_{3.2}$) mean particle size according to eqn (3).³⁰

$$V(h) = \begin{cases} \infty & \text{if } h < \min(fL_i, fL_j) \\ 0 & \text{if } h \geq \min(fL_i, fL_j) \end{cases} \quad (2)$$

$$\sigma^2 = \ln(d_{4.3}/d_{3.2}) \quad (3)$$

2.4 Atomic force microscopy (AFM)

AFM measurements were performed using a Nanowizard II (JPK Instruments, Bruker) mounted on an inverted fluorescent optical microscope. All measurements were performed in the closed loop mode. Indentation curves were recorded using driving speeds ranging from 200 nm s^{-1} to 500 nm s^{-1} to minimize the impact of hydrodynamic drag that at all times was $< 5 \text{ \AA}$ in the deflection equivalent.³¹ DNP-10 Si_3N_4 probes (AFM Probes, Bruker) with tip radius $\sim 20 \text{ nm}$ were used for both imaging and force volume spectroscopy/force mapping. Immediately before use, probes were cleaned in oxygen plasma for 5 min and then were mounted and immersed in the experimental cuvette with buffer. The spring constant (k_z) of the sensors ranged from 0.3 N m^{-1} to 0.5 N m^{-1} and was determined using the Asylum Research GetReal™ routine that utilizes a combination of the thermal noise and the Sader calibration methods.³² Each cantilever was first mounted on the Cypher AFM (Asylum Research, Oxford Instruments) to perform calibrations and then transferred to the JPK instrument, where the resonance frequency and thermal noise spectrum were double-checked before performing the force mapping experiments.

The AFM probe is used to perform nano-indentation with the reduced modulus of the particle, $E_{rp} = \frac{E_p}{1 - \nu_p^2}$, (E_p and ν_p are the Young's modulus and Poisson's ratio of the particle, respectively) extracted from force-distance curves using the multi-regime analysis (MRA) method described previously in Bonilla *et al.*³³ Briefly, the force response of a microgel particle is described using a series of resistors, which represent different types of material deformation. The resulting field of force vectors

Table 1 Particle modulus from single particle AFM (average value) (E_p) and bulk gel modulus (G_p); k values found using Batchelor's model; and particle size distribution for 5 wt% and 1.5 wt% agarose microgels

Agar %	E_p (kPa)	G_p (kPa)	k (dl g^{-1})	$d_{0.5}$ (μm)	$d_{4.3}$ (μm)	$d_{3.2}$ (μm)	σ	ϕ_{rcp}	ϕ_j
5	399.2 ± 41.5	234	1.92	4.88	5.28	4.43	0.44	0.69	0.72
1.5	21.7 ± 1.6	64	7.30	3.09	19.36	2.71	0.52	0.86	0.93

is mapped onto the deformation space to predict the total indentation (deflection) measured by AFM. In the current analysis, three mechanical deformation models are considered as follows:

(1) The empirical polymer brush compression model, which describes the force response associated with the outer layer of hydrogel particles that appears to behave as a dense polymer brush.

(2) Hertzian continuum mechanics model, which describes small scale contact deformation of microgel and is used to determine E_{rp} .

(3) Ogden hyper-elastic model, which describes larger-scale non-linear deformations.

In a typical experiment, the indentations exerted on a particle are kept low to ensure that non-linear effects are minimised. The inclusion of the Ogden hyper-elastic and the bush-like compression models provides a reliable method of determining the upper and lower bounds of linear deformations, respectively. In essence, the fitting procedure filters-out all non-Hertzian contributions and ensures that only Hertzian-like deformations are considered in the evaluation of the elastic modulus.

The AFM nanoindentation data on microgels was collected using force-volume (FV) techniques, which enables mapping the topography, determining the shape of surface bound microgels and performing nanoindentation measurements in a single experiment. Through FV imaging, we locate the particle apex where mechanical measurements can be interpreted using simple assumptions; a microgel sphere is co-axially indented by a conical tip with a spherical cap. The sphere to cone transition is described using the parametric form developed by Briscoe *et al.*³⁴

$$F = 2E_{rp} \left\{ a\delta - \frac{a^2}{2 \tan \theta} \left[\frac{\pi}{2} - \sin^{-1} \left(\frac{b}{a} \right) \right] - \frac{a^3}{3R_{tip}} + \sqrt{a^2 - b^2} \left[\frac{b}{2 \tan \theta} + \frac{a^2 - b^2}{3R_{tip}} \right] \right\} \quad (4)$$

F is the force, δ is the maximum indentation, θ is the half-angle of the cone, R_{tip} is the radius of AFM tip (spherical cap). The radius of contact a is defined using eqn (5):

$$\delta - \frac{a}{R_{tip}} \left[\sqrt{a^2 - b^2} - a \right] - \frac{a}{\tan \theta} \left[\frac{\pi}{2} - \sin^{-1} \left(\frac{b}{a} \right) \right] = 0 \quad (5)$$

The transition point b for the case of a spherical tip merging smoothly with the body of the cone is defined as: $b = R_{tip} \cos \theta$.

The elastic modulus was extracted from the curves collected near the apex of each particle tested. Due to their weak negative charge and hydration, agarose microgels do not spontaneously adhere to the glass or mica substrate. We therefore utilised poly-L-lysine priming layer to render the glass surface positively charged and enable the microgels to weakly adhere to it. Importantly, the adhesive force is found to be small, and particles can be easily displaced from the surface; this ensures that adhesion induced particle deformation can be ignored. The weak adhesion allows particle slip under cantilever

pressure; to avoid this we have utilised very weak 3–5 nN forces to perform FV imaging, and only when at the apex we recorded force curves with the larger indentation set points of 10–20 nN to enable determination of the modulus. At the apex, the tip force acts in a direction perpendicular to that of the substrate, hence sliding is generally avoided.

3. Results and discussion

3.1 Suspension rheology below RCP: viscous flow regime

Our previous study presented rheological flow curves for agarose microgel suspensions (at polymer concentrations of 0.5, 1.5, 2.5, 4, & 5 wt%) and demonstrated the validity of the modified-MPQ model to describe the viscosity up to ϕ_{RCP} .¹³ Here, we focus on microgels containing 5 wt% and 1.5 wt% agarose, where the measured particle size distributions were used in the Farr and Groot¹¹ analytical model to predict $\phi_{RCP} = 0.69$ and $\phi_{RCP} = 0.86$ respectively (note, Table 1 provides relevant size data for the model; and size distributions can be found in ESI†). The inverted viscosity term ($1/\sqrt{\eta_r}$) for these two suspensions is shown in Fig. 1, with data only included that had flow curves that classified the suspensions as Newtonian or near-Newtonian ($0.9 < n < 1.1$, according to Power Law model) for $\phi_0/\phi_{RCP} < 1$, with the exception of 1.5 wt% agarose suspension at $\phi_0 = 0.66$ that had a power law index of $n = 0.83$. The viscosity data for both sets of suspensions is linear when presented in this form, and closely matches the independent prediction provided by the modified-MPQ model (eqn (1)). This shows that the differences between the average viscosity of suspensions

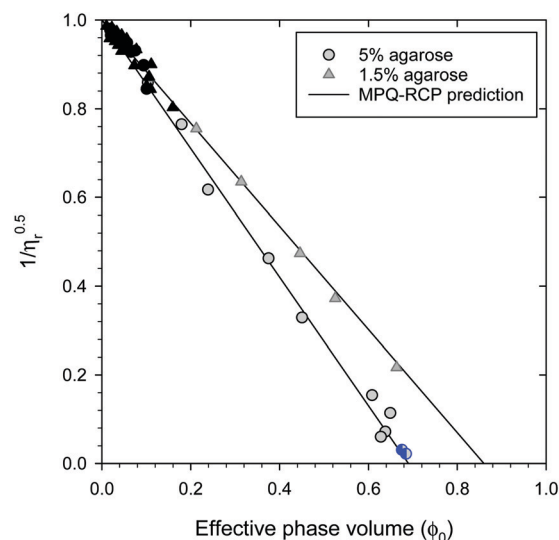


Fig. 1 Relative viscosity presented as $1/\sqrt{\eta_r}$ against effective phase volume measured for 1.5 wt% and 5 wt% agarose microgel suspensions and below random close packing ($\phi_{RCP} = 0.86$ and 0.69 , respectively). Of the data included, grey-filled symbols are used to indicate the average viscosity specifically shown in Fig. 2, and data is only used if $n > 0.9$ when fitted to a power law model. This is also the case for the blue-grey symbols, which are used to indicate the suspensions also exhibit viscoelastic fluid-like response. The lines are predictions using the MPQ-rpc model.

containing 5 wt% agarose and 1.5 wt% agarose microgels is principally due to their different particle size distributions.

The flow curves of each set of suspensions for $\phi_0 < \phi_{\text{RCP}}$ are included in Fig. 2a and 3a (see grey and mixed grey/blue symbols), with the average viscosity values for Newtonian or near-Newtonian fluids indicated by solid-lines. The slight non-linearity in flow curves may arise from particle migration effects during measurement, but the measurements are consistent with other studies on similar microgels.^{13,19} Key points to note include the observation of a slight degree of shear thickening when $\phi_0 \geq 0.44$ for the 5 wt% agarose microgels (Fig. 2a); this is considered to be due to excluded volume effects influencing the behaviour of non-Brownian particles.³⁵ Shear-thickening effects are not observed for the 1.5 wt% agarose microgels; we have noted previously that such effects occur for harder microgels.¹³

We also note for the 1.5 wt% agarose (Fig. 3a) that classical power-law shear-thinning occurs for $\phi_0 = 0.66$ ($n = 0.83$) and $\phi_0 = 0.78$ ($n = 0.53$) that corresponds to $\phi_0/\phi_{\text{RCP}} = 0.77$ and $\phi_0/\phi_{\text{RCP}} = 0.91$, respectively, and that the Herschel-Bulkley model best describes the flow curve at $\phi_0 = 0.83$ ($\phi_0/\phi_{\text{RCP}} = 0.97$) albeit with an almost negligible yield stress ($\sigma_y = 1$ Pa). Hence, the 1.5 wt% agarose microgel suspension exhibits shear-thinning well below ϕ_{RCP} , while significant shear-thinning only occurs above ϕ_{RCP} for the 5 wt% agarose microgels.

In addition to shear viscosity for the samples with $\phi_0 < \phi_{\text{RCP}}$, all the suspensions with $\phi_0/\phi_{\text{RCP}} < 0.97$ do not have measurable G' values, as shown in Fig. 2b and 3b. The three samples in the region $0.97 < \phi_0/\phi_{\text{RCP}} < 1$ with measurable linear viscoelasticity all exhibit the response expected for 2nd order viscoelastic fluids, whereby $G' \sim \omega^2$ and $G'' \sim \omega$ (see blue/grey data). Given the experimental error in phase volume, sizing and rheology measurements, this transition to

a viscoelastic fluid occurs sufficiently close to ϕ_{RCP} to reasonably consider that a defining criterion based on current data. We thus conclude that agarose microgel suspensions are viscous for $\phi_0 < \phi_{\text{RCP}}$, whereby viscous behaviour suggests that particles move past their nearest neighbours within the experimental time scale.³⁶ Thus, as particles pack close together at ϕ_{RCP} , which is dependent on the particle size distribution, the suspension undergoes a viscous-to-viscoelastic fluid transition. We hypothesise that random close packing is the point at which viscoelasticity is first apparent, which arises because of the proximity of particles inducing a significant relaxation time scale prior to their transition to a jammed viscoelastic solid-like state on further increases in phase volume.

3.2 Suspension rheology at and above random close packing: viscoelastic regimes

Fig. 2 and 3 show the flow curves and linear viscoelastic frequency sweeps for each set of microgel suspensions at and above the phase volumes corresponding to random close packing. Blue symbols are used to indicate a viscoelastic fluid response ($G' < G''$) while black symbols are used to indicate viscoelastic solid-like response ($G' > G''$). All samples where $\phi_0 > \phi_{\text{RCP}}$ are characterised by the Herschel-Bulkley fluid model, with yield stresses ranging from 25 to 320 Pa.

The microgel suspensions are found to transition to a viscoelastic solid-like rheology upon increasing phase volumes just above ϕ_{RCP} , where $G' \approx G''$ over a decade in frequency at $\phi_0 = 0.70$ ($\phi_0/\phi_{\text{RCP}} = 1.01$) for 5 wt% agarose microgels (Fig. 2b) and above $\phi_0 = 0.92$ ($\phi_0/\phi_{\text{RCP}} = 1.07$) for 1.5 wt% agarose microgels (Fig. 3b). Above these volume fractions, G' is essentially constant with frequency in the measured region and greater than G'' while the suspensions also exhibit a yield stress, which indicates that the suspensions are viscoelastic

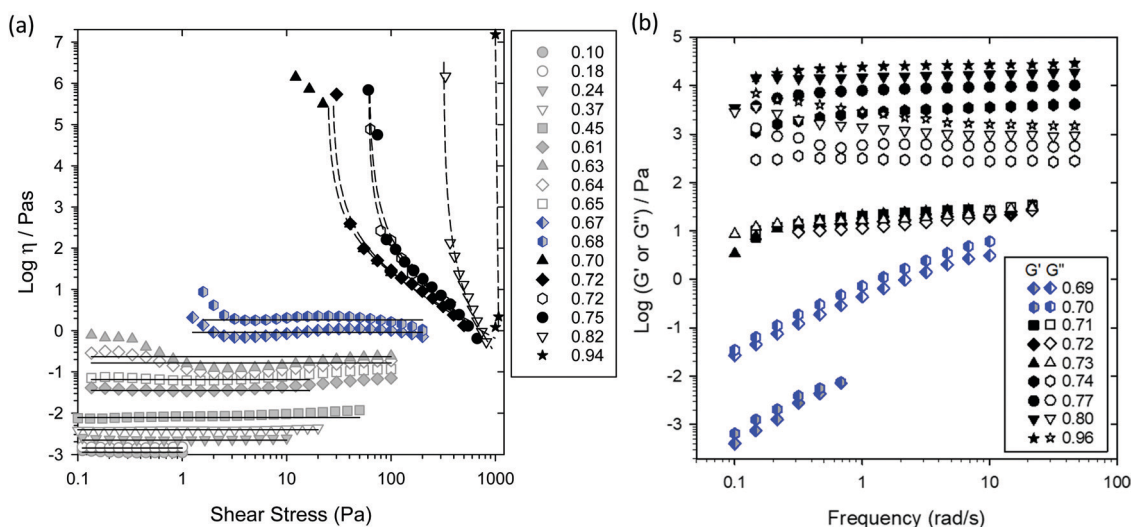


Fig. 2 Rheology of suspensions of hydrogel microspheres containing 5 wt% agarose, showing (a) shear rheology (alternating filled and empty symbols are used to guide the eye) and (b) linear viscoelasticity (filled black or blue-grey data is G' ; empty or blue-white data is G''). The legends show the effective phase volume (ϕ_0). Grey data corresponds to $\phi < \phi_{\text{RCP}} = 0.69$ (viscous); blue data corresponds to where measurable $G' < G''$ (viscoelastic fluid); black data corresponds to where $G' > G''$ at 10 rad s^{-1} (viscoelastic solid). Half blue symbols indicate that $\phi < \phi_{\text{RCP}}$ and viscoelastic fluid. Solid lines show the region of average viscosity for viscous fluids, and dashed lines are fits to the Herschel–Bulkley model.

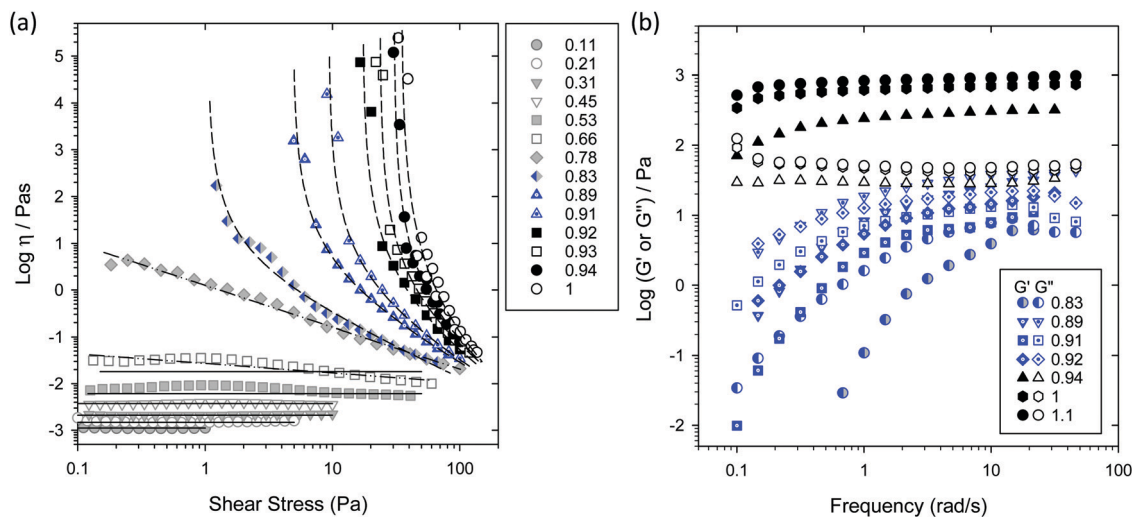


Fig. 3 Rheology of suspensions of hydrogel microspheres containing 1.5 wt% agarose, showing (a) shear rheology (alternating filled and empty symbols are used to guide the eye) and (b) linear viscoelasticity (filled black or blue-grey data is G' ; empty or blue-white data is G''). The legends show the effective phase volume (ϕ_0). Grey data corresponds to $\phi < \phi_{rcp} = 0.86$ (viscous); blue data corresponds to where measurable $G' < G''$ (viscoelastic liquid); black data corresponds to where $G' > G''$ at 1 rad s^{-1} (viscoelastic solid). Half blue symbols indicate to $\phi < \phi_{rcp}$ and viscoelastic fluid. Solid lines show the region of average viscosity for viscous near-Newtonian fluids, dash-dot lines are fits to power law, and dashed lines are fits to the Herschel-Bulkley model.

soft solids whereby the microstructure is in a jammed state. Considering that the moduli are relatively independent of frequency in this domain, we consider the phase volume at which the $G' = G''$ at a set frequency (arbitrarily, 10 rad s^{-1}) to indicate the critical point at which suspensions transition to a jammed state, which is denoted by ϕ_j . Note that for 1.5 wt% microgels, values of $\phi_0 > 1$ suggest that soft microgels are no longer fully swollen above ϕ_j where close contact is occurring.

It is common to define the LST to be when the $\tan \delta = 1$ at a specified frequency (e.g. 1 or 10 rad s^{-1}). But while this is a convenient method, its accuracy is considered limited due to the frequency dependency of G' and G'' . An alternative criterion was proposed by Winter and Chambon,³⁷ who suggest that the rheology at LST is governed by self-similar relaxation, as indicated by $\tan \delta$ being independent of frequency. This definition suggests that the microgel suspensions here are at the LST when $\phi_0 = \phi_{rcp}$. However, the structure is fragile at this point, as noted by the frequency dependence of the moduli, whereby particle rearrangements are still occurring on a measurable timescale.

The substantial difference in actual phase volumes for LST between the two microgels is significant, which may arise due to 1.5 wt% agarose microgels being softer than 5 wt% agarose ones, as well as due to the major difference in size distribution that define how they pack together. We also see a viscoelastic fluid response occurs over a wider phase volume range for the softer microgels, and while this may result from its softness (micro-mechanics), this may also be due to its wider particle size distribution.

3.3 Micromechanics

3.3.1 Theory of close packed elastic spheres. When elastic particles are in contact, the linear viscoelastic response is very much dependent on the individual particle modulus, the separation distance between neighbouring particles, and the microstructure.¹⁹ Particle deformation gives rise to strong

repulsive interactions, and this is captured from statistical mechanics using the Zwanzig and Mountain³⁸ expression that gives the high frequency plateau modulus of monodisperse spherical particles as a function of the interaction potential and pair correlation function.¹⁰ In considering this for non-colloidal microgels, Evans and Lips¹⁰ assumed that non-central forces arising from friction and adhesion, which are described by the Johnson, Kendall and Roberts (JKR) theory,³⁹ are negligible and that only nearest neighbours interact. In this case the interaction potential between two elastic spheres in contact is a function of separation distance, which is described through Hertzian contact mechanics as given by eqn (6) in terms of particle radius (R_0), centre to centre separation (r) and the particle reduced elastic modulus $E_{rp} = \frac{E_p}{1 - \nu_p^2}$, where E_p and ν_p are particles' Young's modulus and Poisson's ratio, respectively.

$$V(r) = 16/15 E_{rp} R_0^{1/2} (R_0 - (r/2))^{2.5} \quad (6)$$

The Evans and Lips model,¹⁰ as given by eqn (7), is used to predict the evolution of the modulus of our microgel suspensions with increasing phase volume. A relative phase volume ($\phi_r = \phi_0 / \phi_j$) is used to describe packing fraction relative to the phase volume at which contact occurs, which we define to be ϕ_j . We use the number of nearest neighbours (N) determined by Rognon *et al.*²⁶ through discrete element simulation of packed suspensions of non-colloidal elastic spheres ($N = 10$).

$$G = E_{rp} \left[\phi_r^{1/3} \left(1 - \phi_r^{-1/3} \right)^{0.5} - \left(\frac{8}{3} \right) \phi_r^{2/3} \left(1 - \phi_r^{-1/3} \right)^{1.5} \right] \left(\frac{\phi_j N}{10\pi} \right) \quad (7)$$

For homogeneous isotropic materials, the particle shear modulus (G_p) may be used by substituting E_{rp} for $G_p/2(1 - \nu)$.

3.3.2 Nanoindentation. A key challenge when utilising the Evans and Lips model is being able to measure the mechanical properties of individual spheres to yield E_{rp} . To do this, we utilise AFM nanoindentation to measure the mechanical response of a large number of microgels, and extract E_{rp} from the Hertzian part of the nanoindentation curve using multi-regime analysis routine described briefly above as well as in Bonilla *et al.*³³ The derivation of the Evans and Lips model is based on Hertz' model of contact mechanics and therefore requires knowledge of the reduced modulus, $E_{rp} = \frac{E_p}{1 - \nu_p^2}$. The advantage of using AFM is that it conveniently measures E_{rp} directly and therefore no assumptions are required with regards to the value of the Poisson's ratio. We note that contributions from the deformations of a silicon nitride tip ($E \approx 200$ GPa) are truly negligible:

$$\langle E_r^* \rangle_{\text{AFM}} = E_{rp} \cdot \left(1 - O\left(\frac{E_{rp}}{E_{\text{tip}}}\right) \right) \equiv E_{rp} \cdot (1 - O(10^{-6})) \quad (8)$$

Fig. 4(a and b) show the FV image of the particles and the corresponding bright field image, respectively. One can see that particle height is approximately equal to its diameter, which supports the assumption that adhesion induced deformation is small. This assumption is crucial; the adhesion induced deformation causes stress in the particles, which affects interpretation of indentation results and evaluation of the modulus. Fig. 4(c) shows two sets of typical force-indentation curves recorded on the 5 wt% and 1.5 wt% agarose particles. The applicability of the Hertzian model is confined to a small range of indentations, where the dependency of central indentation distance on the

contact area is linear in accordance with Hertzian approximation, which justifies the use of multi-regime analysis (MRA) routine for data analysis. The analysis across 12 individual microgel particles (10–40 force curves per particle) yielded the following values of $E_{rp} = 21.7 \pm 1.6$ kPa and 399.2 ± 41.5 kPa for 1.5 wt% and 5.0 wt% agarose gels, respectively, as shown in Table 1.

3.4 Regime map of elastic particle suspensions

Fig. 5a and b show η_{r0} , as well as G' and G'' at 10 rad s^{-1} , as a function of ϕ_0/ϕ_{rcp} for 5 wt% and 1.5 wt% agarose microgels respectively. Plotting these properties together highlights the presence of four distinct regimes described as follows upon increasing phase volume:

(I) Dilute Newtonian regime $\phi_0 < 0.05$, where Einstein's equation for hard spheres applies ($\eta_r = 1 + 2.5\phi_0$).

(II) Viscous regime, up to ϕ_{rcp} : there is no apparent viscoelasticity and the viscosity is predicted by the modified-MPQ model.

(III) Viscoelastic fluid regime, $\phi_{\text{rcp}} < \phi_0 < \phi_j$, where suspensions are in transition between a liquid and solid.

(IV) Viscoelastic solid regime, $\phi_0 \geq \phi_j$, where the suspension has a jammed structure, the moduli are independent of frequency and $G' > G''$, and shear rheology is defined by a yield stress.

Regimes II to IV, illustrated in Fig. 5a and b, are reminiscent of two rather dissimilar families of systems: (i) colloidal suspensions and (ii) frictional-adhesive non-colloidal suspensions. In system (i), the suspension experiences an entropic glass transition at $\phi_g < \phi_j$ when particles become trapped in metastable cages restricting long-range diffusion beyond

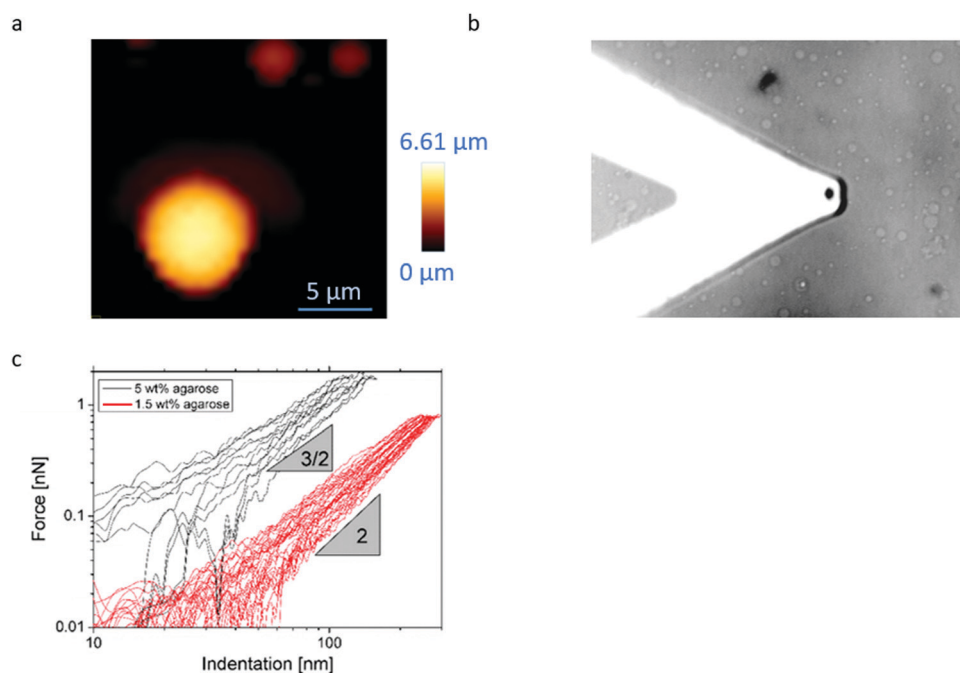


Fig. 4 (a) FV height image of a typical 5 wt% agarose particle. (b) The corresponding bright field image. (c) Typical force-indentation curves for 5 wt% and 1.5 wt% agarose particles plotted in logarithmic coordinates.

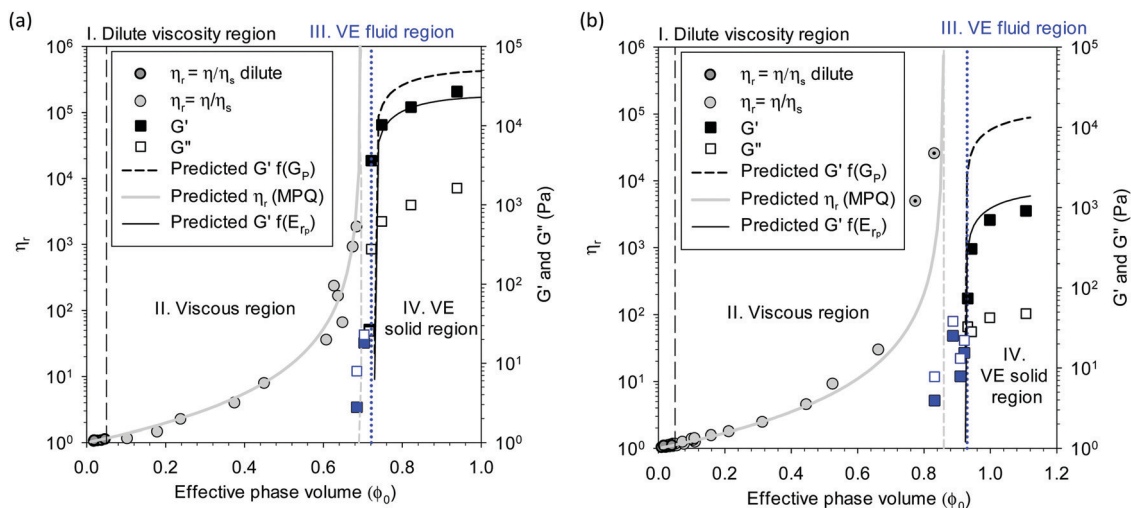


Fig. 5 Relative viscosity and suspension G' and G'' (at $\omega = 10 \text{ rad s}^{-1}$) against ϕ_0 for (a) 5 wt% agarose and (b) 1.5 wt% agarose microgel suspensions. Dotted symbols indicate fluids were shear-thinning ($n < 0.9$) and the consistency index is used in place of viscosity. Vertical lines indicate boundaries for each region: Black long dash line is $\phi_0 = 0.05$, grey-dashed line is $\phi_0 = \phi_{\text{rcp}}$ and blue-dotted line is $\phi_0 = \phi_j$ (jamming transition). Different regions of behaviour are indicated as (I) Dilute Newtonian viscosity regime $\phi_0 < 0.05$ (II) viscous, $\phi_0 < \phi_{\text{rcp}}$ (III) viscoelastic fluid, $\phi_{\text{rcp}} \leq \phi_0 \leq \phi_j$, and (IV) viscoelastic solid $\phi_0 > \phi_j$.

experimental time scales. Solid-like behaviour arises due to elastic restoring forces of entropic origin, driving the particles back to their metastable equilibrium position.^{6,40} While this behaviour is shared by both soft and hard colloidal suspensions, the specific micromechanics beyond ϕ_j differs significantly in either case.⁴¹ Polydisperse suspensions can contain simultaneously colloidal and non-colloidal particles and thus may exhibit the double glass/jamming transitions.⁴² This does not appear to be the case for our system, given that the particle size is well above typical colloidal sizes (see ESI,† Fig. S1). Most importantly, the first transition occurs at the estimated ϕ_{rcp} , while a possible glass transition should theoretically manifest at $\phi_g < \phi_{\text{rcp}}$ (e.g. $\phi_g \approx 0.58$ for both monodisperse hard and soft colloidal suspensions, if deswelling and compressive effects are accounted for in the latter case).^{6,43}

More intriguing is the possibility of case (ii): frictional-adhesive non-colloidal suspensions. The maximum amorphous packing for frictionless, adhesionless hard spheres is random close packing ($\phi_{\text{rcp}} = 0.64$ for monodisperse systems). However, interparticle friction can lead to the onset of solid-like behaviour at a random loose packing $\phi_{\text{rlp}} < \phi_{\text{rcp}}$, so long as the steady state shear stress is above a certain critical value. The increase in friction with increasing roughness of hard particle suspensions at the same phase volume is shown by Hoyle *et al.*⁴⁴ Upon comparison of glass spheres⁴⁴ with polystyrene spheres⁴⁵ they also suggest that the degree of viscosity increase is a function of particle friction coefficient. At the limit of infinite friction coefficient between particles, the random loose packing fraction can be theoretically reduced to 0.55.^{2,46,47} Shear thickening in frictional suspensions (such as corn-starch in aqueous media) is currently thought to be fully or partially driven by the formation of compressive frictional contacts between neighbouring particles beyond the critical

stress required overcomes the stabilizing interparticle repulsion.⁴⁸ Nonetheless, viscosity divergence occurs at random loose packing, at ϕ_{rlp} , while in this work it is found at ϕ_{rcp} . And evidently, we do not observe discontinuous shear thickening effects.⁴⁴

In our system the adhesion is found to be small, whilst the repulsive barrier is equally weak, resulting in the scenario that is distinct from the two cases described above. Whilst the word “adhesive” is often understood as “attractive” and the word “repulsive” often connotes “non-adhesive”, in our case it is more pertinent to use the definition of the adhesive contact introduced by Richards *et al.*:⁴⁹ it denotes a finite contact area with a concomitant rolling constraint. In the absence of the strong repulsive barrier, the formation of frictional contact is underpinned by material deformation and the interlocking mechanism, whereby the interplay between rolling and sliding at the particle contact points can lead to a double transition, analogous to that observed in colloidal systems.^{6,43} Under continuous flow, it has been shown that such a system may exhibit solid-like behaviour at $\phi < \phi_{\text{rlp}}$ driven by rolling constraint, showing a finite yield stress σ_y and shear thinning flow at $\sigma > \sigma_y$. Conversely, at $\phi_{\text{rlp}} < \phi_{\text{rcp}}$ a friction-driven diverging yield stress is observed.^{49,50} This double transition was reported for rigid granules (corn starch in oil) and it is reasonable to expect the transition volume fractions could be much higher in soft microgel systems, as the particles can deform. Interestingly, oscillatory flow removes frictional contacts, and consequently the system does not jam until ϕ_{rcp} , *i.e.*, the applied oscillatory shear is sufficient to overcome the rolling constraint and mobilize the contact.

The rolling constrained described above may originate from the adhesion mechanism or the physical interlocking mechanism. The key difference our suspensions displays in comparison with

the adhesive-frictional suspensions is that the first transition occurs at $\phi_{\text{rcp}} < \phi_j$. However, the combination of softness and polydispersity may significantly alter the volume fractions for the corresponding transitions. We will advance a potential theory in the concluding remarks.

We use the Evans and Lips model to predict G' in the viscoelastic solid regime. Included in Fig. 5a and b are two predicted curves that only differ in the experimental data used to define E_{rp} in eqn (7). Firstly, we assume the microgels have equivalent viscoelasticity to bulk gels (G_{p}) made at the same nominal polymer concentration. We did this because it has been used by ourselves and others previously due to a lack of data on individual particle mechanics. The second predicted curve (solid black line) uses the average E_{rp} measured directly using the AFM on individual microgels.

The predicted curve using E_{rp} accurately predicts the suspension modulus for both 5 and 1.5 wt% agarose suspensions. In contrast, the prediction found using G_{p} overestimates the suspension modulus by up to an order of magnitude for the 1.5 wt% agarose microgels. We can positively conclude that the Evans and Lips model predicts the steep rise in modulus at our defined value of ϕ_j (i.e. ϕ_0 when $G'/G'' = 1$ at frequency of 10 rad s^{-1}). The model provides a reasonable prediction of the measured data, with a directly measured value of E_{rp} . This supports our hypothesis that bulk gel modulus does not provide an accurate estimate for particle modulus, particularly where polymer concentrations are not identical to that of the bulk gel or where materials are non-homogeneous non-isotropic materials as is the case for agarose. This result also suggests that the model for Hertzian deformation of these non-Brownian microgels is relevant to suspensions at phase volumes well above RCP. Recent modelling work by Rovigatti *et al.*⁵¹ suggests that neutral (non-ionic) Brownian microgels behave as Hertzian spheres up to close packed conditions whereupon it was speculated that this may breakdown due to the debatable use of pair-potentials in the jammed (concentrated) regime, and the potential for faceting, interpenetration, and interparticle entanglement.^{5,51–53} While it is challenging to directly measure the mechanics of Brownian microgels, the data provided here may be used to consider

several of these effects that are still relevant to non-Brownian spheres to ascertain whether they could improve the prediction of the suspension modulus, particularly for the more polydisperse and softer microgels.

4. Conclusions

Experimental evidence is shown here for the existence of a viscoelastic fluid region at the LST in the vicinity of ϕ_{rcp} , which has not previously been observed experimentally. This experimental evidence is supported using existing models for the viscosity-phase volume and suspension modulus-phase volume without using free-fitting parameters. Four distinct regions are observed with boundaries defined at: $\phi_0 < 0.05$, above which particle interaction begin to occur; using ϕ_{rcp} , which is obtained using the particle size distribution; and the crossover of G' and G'' to give ϕ_j . Below ϕ_{rcp} , the suspension viscosity depends only on phase volume and particle size distribution. At ϕ_{rcp} , the suspension is in transition between liquid and solid but is not in a jammed state. The jammed state is entered at values of ϕ_j that are above ϕ_{rcp} , and the evolution of suspension modulus is reasonably described by the Evans and Lips model. A key finding here is that the suspensions are only in a jammed solid-like state when $\tan \delta = 1$, which is supported by our observation that this also corresponds to order of magnitude increases in moduli, as predicted by the Evans and Lips model that incorporates Hertzian contact mechanics.

Another key observation is that ϕ_{rcp} appears to correspond to the start of a LST, as indicated by G' and G'' having self-similar responses with frequency (constant $\tan \delta$), but is also expected from a microstructural point of view if the structure is randomly close packed whereby the viscosity asymptotes towards infinite values. However, while microgels may be very weakly contacting, the suspensions are still fluid-like with $G' < G''$ and the overall modulus is low.

Future work is needed to probe the mechanism that dictates the phase volume range across which the viscoelastic fluid behaviour occurs. The viscoelastic fluid behaviour observed at $\phi_{\text{rcp}} \leq \phi_0 \leq \phi_j$ is significantly wider for the softer 1.5 wt%

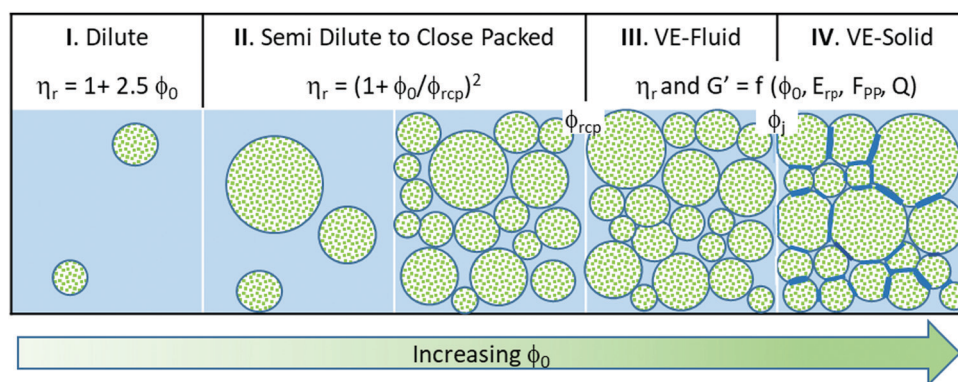


Fig. 6 The 4 regimes of soft particle suspension rheology as a function of ϕ_0 . Up to ϕ_{rcp} , $\eta_r = f(\phi_0)$. In the viscoelastic regime, G' is: fluid-like, where inter-particle friction (F_{pp}) and poroelastic permittivity of the medium (Q) dominate; or solid-like where E_{p} is dominant.

agarose microgels. We hypothesise that the width of the transition region is dictated by the properties of the frictional contact, particle size distribution, surface roughness and particle modulus. Impact of particle roughness and polydispersity has been extensively studied and modelled for hard sphere suspensions up to maximum packing fraction, as comprehensively reviewed by Tanner (2019).⁵⁴ However, further work is needed to probe these effects in soft particle suspensions above ϕ_{rec} . All four parameters individually and collectively are likely to directly impact friction between particles, which in this regime cannot be neglected. With increasing ϕ an increasing number of particles become trapped encompassing progressively larger particle size range. In other words, with increasing ϕ , the larger proportion of particles from the lower end of the particle size distribution lose the ability to move freely within the assembly that at ϕ_{rec} is dominated by the largest of particles. This effect may also be responsible for the non-shear-thinning occurring for the 1.5 wt% agarose microgels below the actual value of ϕ_{rec} . With increasing ϕ , the particle compression also increases, which results in the rise of the inter-particle friction force according to the relationship in eqn (9), which relates the dissipative friction force (F_f) with the conservative mechanical spring force (F_N):

$$F_f = \mu F_N = \frac{2}{3} \mu E_r R_0^2 \left(1 - \Phi_{ij}^{-1/3}\right)^{1.5} \quad (9)$$

μ is the friction coefficient between contacting spheres. At the certain value of ϕ , F_f becomes large enough to set a condition where it is energetically easier to deform particles (and form a fully jammed suspension) than for particles to slide past each other to sustain suspension fluidity. At the point of jamming, the suspension behaves elastically. Another important factor is that microgels are porous and hence subject to poro-viscoelastic compression. When soft microgels are compressed the suspension phase volume increases beyond that predicted by the random close packing models, which we illustrate schematically in Fig. 6. Future work should aim to further decouple the influence of particle modulus and particle size distribution.

We anticipate that the observed behaviours are relevant to other suspensions involving deformable entities. In particular, our future work seeks to use this approach to explain the rheology–structure relationship in complex soft material systems that are found in foods and personal care products. These are often soft solid-like materials that rely on being packed suspensions of soft particulate-like entities, and our work here provides a simple framework in which to examine such complex systems.

Conflicts of interest

There are no conflicts to declare.

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